Mark Scheme (Results)

Mock Set 1

Pearson Edexcel GCE
In AS Mathematics (8MA0) Paper 21

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General Marking Guidance

- All candidates must receive the same treatment. Examiners must mark the first candidate in exactly the same way as they mark the last.
- Mark schemes should be applied positively. Candidates must be rewarded for what they have shown they can do rather than penalised for omissions.
- Examiners should mark according to the mark scheme not according to their perception of where the grade boundaries may lie.
- There is no ceiling on achievement. All marks on the mark scheme should be used appropriately.
- All the marks on the mark scheme are designed to be awarded. Examiners should always award full marks if deserved, i.e. if the answer matches the mark scheme. Examiners should also be prepared to award zero marks if the candidate's response is not worthy of credit according to the mark scheme.
- Where some judgement is required, mark schemes will provide the principles by which marks will be awarded and exemplification may be limited.
- When examiners are in doubt regarding the application of the mark scheme to a candidate's response, the team leader must be consulted.
- Crossed out work should be marked UNLESS the candidate has replaced it with an alternative response.

Qu	Scheme	Marks	AOs
1(a)	As (male) elephant shoulder height increases the tusk length increases.	B1	2.4
		(1)	
(b)	Every additional cm of shoulder height gives an additional 0.8[43] cm of tusk length. (o.e.)	B1	3.4
		(1)	
(c)	e.g. he has only used data for male elephants so using this to comment on relationship for all elephants (including female) is not appropriate	B1	2.4
		(1)	
(d)	For finding the area between 75cm and 100cm e.g. each row = 0.1 giving $5\times1.6+10\times1.2+10\times0.8$ or rectangles 8 + 12 + 8 (= 28)	M1	3.1a
	For finding the total area of the histogram e.g. $20\times0.8+10\times1.7+10\times1.2+10\times1.6+10\times1.2+20\times0.8+20\times0.2$ or rectangles 16 + 17 + 12 + 16 + 12 + 16 + 4 (= 93) **and** Percentage between 75cm and 100cm e.g. $\frac{"28"}{"93"}\times100$	M1	1.1b
	30.1…(%)	A1	1.1b
		(3)	
			(6 marks)

Notes:

(a) **B1**: For a correct interpretation of positive correlation, must mention "height" and "length" not just *s* and *t*

(b) **B1**: For a correct interpretation of the gradient of the regression line. Must make reference to rate. Must see value to at least 1 sf

(c) **B1**: A suitable reason e.g. Matthew only used the data for male (or African) elephants

(d) **M1**: For use of the relative heights of the three columns to find a value for area of the histogram between 75cm and 100cm. Ratio of their 1.6, their 1.2 to their 0.8 must be correct i.e. 4:3:2.

M1: For realising the need to find a total area **and** find a percentage. Heights used for the bars must be consistent with those used for the first M mark. If working shown can allow maximum of 2 errors in either the heights or widths in the calculation of the total. An omission is 2 errors.

A1: awrt 30.1(%), allow awrt 30(%) provided full working shown and no errors.

Note: M1M1 may be awarded for $\frac{28}{93}\times100$ oe, but $\frac{28}{93}$ alone with no attempt to obtain a percentage is M1M0

Qu	Scheme	Marks	AOs
2(a)	*A* and *B* labelled correctly or reversed or for *C* labelled correctly together with a reason e.g. since there is no intersection between *A* and *B* or the probability of *A* and *B* happening is zero	B1	2.4
	$(0.08+0.2+0.12)\times(0.12+0.18)[=0.12]$ or $(0.17+0.08)\times(0.08+0.2+0.12)[=0.1]$	M1	1.1b
	$(0.08+0.2+0.12)\times(0.12+0.18)=0.12$ hence independent or $(0.17+0.08)\times(0.08+0.2+0.12)=0.1\neq 0.08$ hence not independent	A1	1.1b
	Correctly deducing labels are *B*, *C*, *A* (in that order)	A1cso	2.2a
		(4)	
(b)	0.75	B1ft	3.4
		(1)	
			(5 marks)

Notes:

(a) **B1:** For *A* and *B* identified as being the two end labels (or *C* identified as the centre label) and correct reason

M1: For a correct calculation to check for independence of the left and middle event or for a correct calculation to check for independence of the middle and right events

You may see: $\frac{0.12}{(0.08+0.2+0.12)}[=0.3]$ or $\frac{0.12}{(0.12+0.18)}[=0.4]$ or $\frac{0.08}{(0.08+0.2+0.12)}[=0.2]$

or $\frac{0.08}{(0.17+0.08)}[=0.32]$ [NB P(*A* | *C*) notation not used since off spec.]

A1: For fully correct working with conclusion of which pair of events are independent (or not)

A1cso: For correct deduction of position of labels *B*, *C*, *A*

Note: If correct working seen for independence condition with labels *B*, *C*, *A* but reason for the mutually exclusive aspect not seen then B0M1A1A1

(b) **B1ft:** ft their *B*

Qu	Scheme	Marks	AOs
3(a)	$\bar{t} = \frac{2876.7}{184} = 15.63...$ $\sigma_t = \sqrt{\frac{46797.3}{184} - \bar{t}^2}$ $\left[= \sqrt{9.903...}\right]$	M1	1.1b
	$= 3.1470...$ awrt 3.15	A1	1.1b
	$28.7 > 15.63... + 3 \times 3.1470...$ [<u>or</u> 28.7 > awrt 25.1]	B1ft	1.1b
		(3)	
(b)	e.g. values range from 0° to 360° as they are angles measured from North, very high or low values would indicate close to North for wind direction rather than being outliers	M1	2.4 [LDS]
	e.g. Wind direction	A1	2.2a [LDS]
		(2)	
(c)	$[2000+]\frac{15}{32} \times 400$ [<u>or</u> $\frac{184}{4} = 46 \therefore$ in $[2000, 2400)$]	M1	2.1
	$= 2187.5$	A1	1.1b
		(2)	
(d)	$\bar{C} = 24.83...$	B1	1.1b
	$\sigma_c = 2.7$	B1	1.1b
		(2)	
		(9 marks)	

Notes:

(a) **M1**: For a correct expression for $\bar{t}$ and either σ_t or s_t ft an incorrect evaluation of $\bar{t}$

Condone missing square root. NB $s_t = \sqrt{9.95783...}$

A1: For $\sigma_t =$ awrt 3.15, may be implied by correct limit for outliers. [$s_t =$ awrt 3.16]

B1ft: For sight of the correct calculation and suitable comparison with 28.7 ft an incorrect evaluation of σ_t or s_t

(b) **M1**: For a suitable reason for not considering outliers

A1: Identifies an appropriate variable

(c) **M1**: Correct fraction $\frac{15}{32} \times 400$ allow a correct equation leading to a correct fraction

e.g. $\frac{x - 2000}{2400 - 2000} = \frac{46 - 31}{63 - 31}$ for M1 Use of $(n+1)$ with 46.25 allow $\frac{15.25}{32} \times 400$

A1: For 2187.5, allow 2190.626 from $(n+1)$ Accept awrt 2190

(d) **B1**: For value of $\bar{C}$ awrt 24.8

B1: For value of σ_c cao

<table>
<tr><th>Qu</th><th>Scheme</th><th>Marks</th><th>AOs</th></tr>
<tr><td rowspan="3">4(a)</td><td>Let A = the number of green beads in a pack $A \sim \mathrm{B}(10, 0.38)$
$\left[\mathrm{P}(A>6)=1-\mathrm{P}(A \leqslant 6)\right]$</td><td>M1</td><td>3.4</td></tr>
<tr><td>$=1-0.9586... =$ awrt 0.0413</td><td>A1</td><td>1.1b</td></tr>
<tr><td></td><td>(2)</td><td></td></tr>
<tr><td rowspan="3">(b)</td><td>
<table>
<tr><td>a</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td></tr>
<tr><td>$\mathrm{P}(A=a)$</td><td>8.392×10^{-3}</td><td>0.0514…</td><td>0.1418…</td><td>0.2318…</td><td>0.2487…</td><td>0.1829…</td></tr>
</table>
<table>
<tr><td>a</td><td>6</td><td>7</td><td>8</td><td>9</td><td>10</td></tr>
<tr><td>$\mathrm{P}(A=a)$</td><td>0.0934…</td><td>0.0327…</td><td>7.52×10^{-3}</td><td>1.024×10^{-3}</td><td>6.278×10^{-5}</td></tr>
</table>
</td><td>M1</td><td>1.1b</td></tr>
<tr><td>Most likely number of green beads is 4 together with
$\mathrm{P}(A=3)=0.2318..., \mathrm{P}(A=4)=0.2487..., \mathrm{P}(A=5)=0.1829...$</td><td>A1</td><td>2.4</td></tr>
<tr><td></td><td>(2)</td><td></td></tr>
<tr><td rowspan="4">(c)</td><td>$\mathrm{H}_0 : p=0.2 \qquad \mathrm{H}_1 : p>0.2$</td><td>B1</td><td>2.5</td></tr>
<tr><td>Let F = number of faulty bowls $\quad F \sim \mathrm{B}(40, 0.2)$
<u>or</u> $\mathrm{P}(F \leqslant 12)=0.9568...>0.95$ or $\mathrm{P}(F \geqslant 13)=$ awrt $0.043...[<0.05]$</td><td>M1</td><td>3.3</td></tr>
<tr><td>CR: $\{F \geqslant 13\}$</td><td>A1</td><td>1.1b</td></tr>
<tr><td></td><td>(3)</td><td></td></tr>
<tr><td rowspan="2">(d)</td><td>awrt 0.043</td><td>B1ft</td><td>1.1b</td></tr>
<tr><td></td><td>(1)</td><td></td></tr>
<tr><td rowspan="2">(e)</td><td>17 is in the CR, sufficient evidence to <u>support</u> the manager's <u>belief</u> or
<u>proportion</u> of faulty bowls is <u>less than</u> 20%</td><td>B1ft</td><td>2.2b</td></tr>
<tr><td></td><td>(1)</td><td></td></tr>
<tr><td rowspan="2">(f)</td><td>A valid model because e.g. <u>random</u> sampling guarantees <u>independence</u> or <u>constant probability</u></td><td>B1</td><td>3.5a</td></tr>
<tr><td></td><td>(1)</td><td></td></tr>
<tr><td colspan="4" align="right">(10 marks)</td></tr>
</table>

Notes:

(a) **M1**: for selecting and using a suitable model, sight of B(10, 0.38) o.e. in words

Can be implied by $\mathrm{P}(A \leqslant 6)$ = awrt 0.959 or final answer = awrt 0.0413

A1: for awrt 0.0413

(b) M1: for finding the probability of at least 2 different numbers of green beads <u>or</u> $\frac{0.38}{r+1} > \frac{0.62}{10-r}$ o.e.

A1: for 4 & $\mathrm{P}(A=3)=0.2318.., \mathrm{P}(A=4)=0.2487.., \mathrm{P}(A=5)=0.1829..$ (accept 2dp) <u>or</u> $r<3.18$

(c) B1: for correctly stating both hypotheses in terms of p or π

M1: For use of tables of B(40, 0.2) to find probability associated with critical value [$\mathrm{P}(F \leqslant 12)$ = awrt 0.957 or $\mathrm{P}(F \geqslant 13)$ = awrt 0.043 (may be implied by either correct probability or by the correct CR)]

A1: $[40 \geqslant]\ F \geqslant 13$ o.e. e.g. $F > 12$ Allow '13 or more' or 'CR $\geqslant$ 13' Correct ans only M1A1

(d) B1ft: awrt 0.043 (allow awrt 4.3%) or correct ft their one-tailed upper CR from B(40,0.2) to 3s.f.

(e) B1ft: For commenting on the manager's belief, ft their (c)
(f) B1: Binomial is a valid model with a suitable reason. Must see underlined words.

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